

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 2
 Duration: 30 minutes

Semester: Summer 2025
 Full Marks: 15

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1. a)

Address	Data	Address	Data
19	(20) ₁₆	14	(34) ₁₆
18	(B2) ₁₆	13	(FF) ₁₆
17	(15) ₁₆	12	(02) ₁₆
16	(89) ₁₆	11	(00) ₁₆
15	(C7) ₁₆	10	(AB) ₁₆

Starting Add = 12

32 bits data $\rightarrow \frac{32}{8} = 4$ locations

12, 13, 14, 15
 LSB MSB

little endian

You are working with a 64-bit RISC-V machine. Registers X₂₁ and X₂₂ contain the numbers (5)₁₀ and (4)₁₀ respectively. You want to update the contents of X₂₁ using 32 bits of data starting from memory address 12.

a) Show the new contents of register X₂₁ after this operation. Your answer should be in hexadecimal. [4]

Answer:

X₂₁ = 0x C734FF02 $\rightarrow$ 0x00000000C734FF02

Reg. size = 64 bits

32 bit Data
 - Word

b) Complete the instruction below to carry out the operation described in question a. [2]

LW X₂₁, 8(X₂₂)

From $\rightarrow$ mem. To $\rightarrow$ Register Load

2.

Sequence1: addi x6, x0, 1 add x5, x29, x6 slli x5, x5, 3 add x5, x5, x10 sd x28, 0(x5)	Sequence2: slli x5, x29, 3 addi x6, x0, 8 add x5, x5, x6 add x5, x5, x10 sd x28, 0(x5)	Assume that the variable f is assigned to register x28. Assume that A[0] and B[0] are in registers x10 and x11 respectively.
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Show that both sequences compute the same memory address in the 4th instruction to store the value from register X28. Explain why? [4+2]

Answer:

addi x6, x0, 1 ; X₆ = 1

add x5, x29, x6 ; X₅ = X₂₉ + 1

Slli x5, x5, 3 ; X₅ = (X₂₉ + 1) * 8

Add x5, x5, x10 ; X₅ = $\downarrow$ + X₁₀

Slli x5, x29, 3 ; X₅ = X₂₉ * 8

addi x6, x0, 8 ; X₆ = 8

add x5, x5, x6 ; X₅ = (X₂₉ * 8) + 8

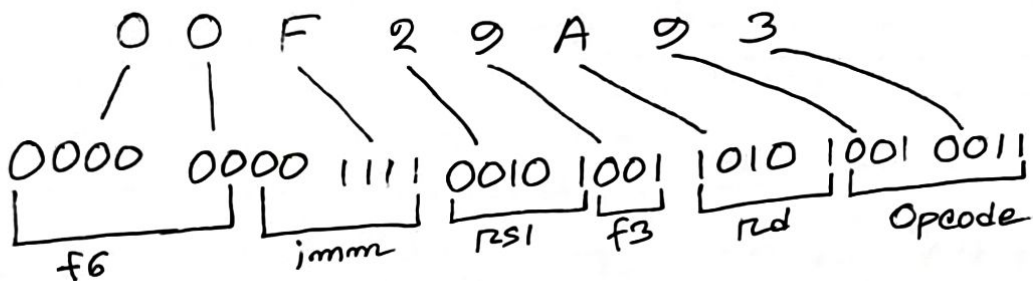
add x5, x5, x10 ; X₅ = $\downarrow$ + X₁₀

$$\begin{array}{l|l}
 X_5 = (X_2 + 1)X_8 & X_5 = (X_2 X_8) + 8 + X_{10} \\
 + X_{10} & \\
 = X_2 X_8 + 8 + X_{10} & = X_2 X_8 + 8 + X_{10}
 \end{array}$$

So, Both are same.

3. Find the RISC-V code of the following machine Codes.
i. 0x00F29A93 [3]

Answer:



Slli X21, X5, 15

Format	Instruction	Opcode	Funct3	Funct7/Funct6
R	ADD	0110011	001	0000000
	SUB	0110011	110	0000001
	AND	0110011	010	0000001
	OR	0110011	011	0000100
	SLL	0110011	101	0000100
I	LD	0000011	000	N/A
	LW	0000011	010	N/A
	LH	0000011	011	N/A
	LB	0000011	001	N/A
	ADDI	0010011	000	N/A
	SLLI	0010011 ✓	001 ✓	0000000
	SRLI	0010011	010	0000000
	ORI	0010011	011	N/A
S	SD	0100011	000	N/A
	SW	0100011	001	N/A
	SH	0100011	010	N/A
	SB	0100011	011	N/A